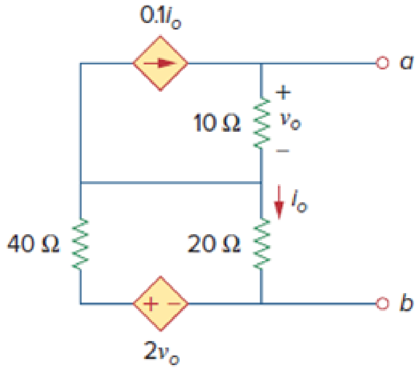
Chapter 4, Problem 62P

Problem

Find the Thevenin equivalent of the circuit in Fig. 4.128.

Figure. 4.128:



Step-by-step solution

Step 1 of 7

Refer to circuit diagram in Figure 4.128 in the textbook.

To find Thevenin's resistance R_{Th}, the independent sources are made equal to zero resistance but the dependent sources are left alone. Because of the presence of the dependent source excite the network with a voltage source v_x connected to the terminals as shown in Figure 1. Let $v_x = 1\text{V}$ to ease our calculations. Find the current i_x through the terminals, and then obtain $R_{\text{Th}} = \frac{1}{i_x}$.

Since, there is no independent source in the circuit, the value of Thevenin's voltage is 0 V.

Therefore, the value of Thevenin's voltage, V_{Th} is $\boxed{0\text{ V}}$.

Step 2 of 7

Draw the circuit diagram with test voltage source.

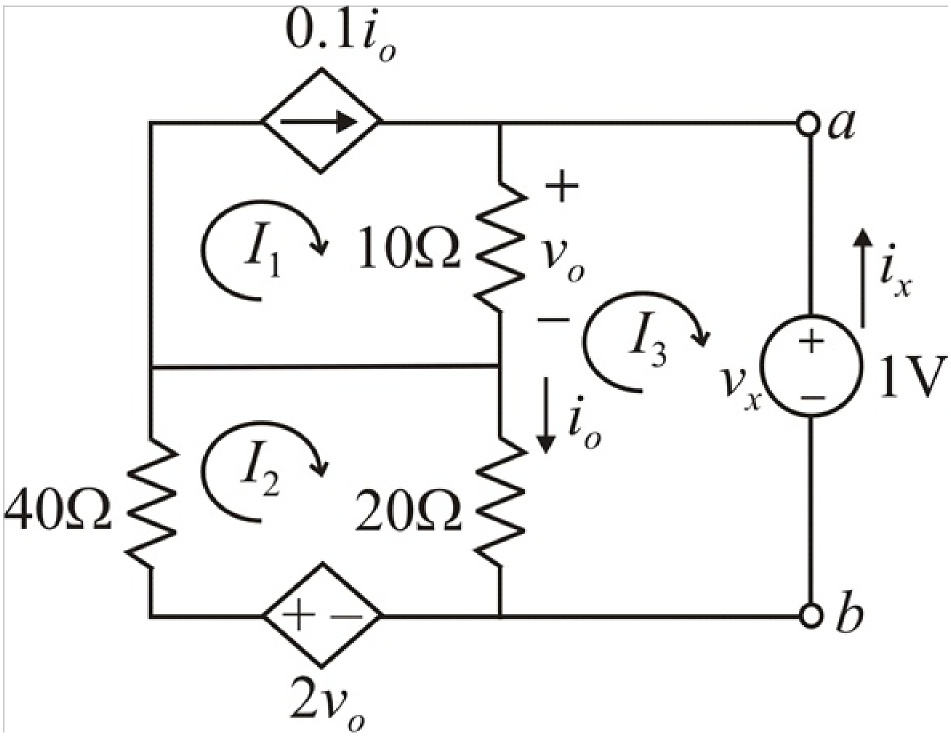


Figure 1

Step 3 of 7

From the circuit shown in Figure 1, the mesh 1 current is,

$I_1 = 0.1i_o$

$i_o = 10I_1$

The current i_o is,

$i_o = I_2 - I_3$

Substitute $10I_1$ for i_o.

$10I_1 = I_2 - I_3$

$10I_1 - I_2 + I_3 = 0$ …… (1)

The voltage v_o is,

$v_o = 10(I_1 - I_3)$

Step 4 of 7

Apply Kirchhoff's Voltage Law to mesh 2.

$$40I_2 + 20(I_2 - I_3) - 2v_o = 0$$

Substitute $10(I_1 - I_3)$ for v_o.

$$40I_2 + 20(I_2 - I_3) - 2\left[10(I_1 - I_3)\right] = 0$$
$$40I_2 + 20I_2 - 20I_3 - 20I_1 + 20I_3 = 0$$
$$-20I_1 + 60I_2 = 0$$
$$20I_1 = 60I_2$$

$I_1 = 3I_2$ …… (2)

Recall equation (1).

$$10I_1 - I_2 + I_3 = 0$$

Substitute $3I_2$ for I_1.

$$10(3I_2) - I_2 + I_3 = 0$$
$$29I_2 = -I_3$$

$I_2 = -\frac{I_3}{29}$ …… (3)

Step 5 of 7

Apply Kirchhoff's Voltage Law to mesh 3.

$$20(I_3 - I_2) + 10(I_3 - I_1) + 1 = 0$$
$$-10I_1 - 20I_2 + 30I_3 = -1$$

Substitute $3I_2$ for I_1.

$$-10(3I_2) - 20I_2 + 30I_3 = -1$$
$$-50I_2 + 30I_3 = -1$$

Substitute $-\frac{I_3}{29}$ for I_2.

$$-50\left(-\frac{I_3}{29}\right) + 30I_3 = -1$$
$$1.724I_3 + 30I_3 = -1$$
$$31.724I_3 = -1$$
$$I_3 = -31.52 \text{ mA}$$

Step 6 of 7

Calculate the current i_x.

$$i_x = -I_3$$

Substitute -31.52 mA for I_3.

$$\begin{aligned} i_x &= -(-31.52 \text{ mA}) \\ &= 31.52 \text{ mA} \end{aligned}$$

Calculate the Thevenin's resistance.

$$\begin{aligned} R_{\text{Th}} &= \frac{1 \text{ V}}{i_x} \\ &= \frac{1 \text{ V}}{31.52 \text{ mA}} \\ &= 31.72\ \Omega \end{aligned}$$

Therefore, the Thevenin's resistance, R_{Th} is $\boxed{31.72\ \Omega}$.

The Thevenin's equivalent circuit is shown in Figure 2.

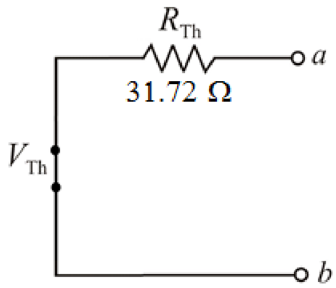


Figure 2

Therefore, the Thevenin's equivalent circuit is shown in Figure 2.